

Chedam: work log

Cumulative record of every working day, newest first. This copy covers up to Tuesday, October 6, 2026. Owner: Sreya.
Baseline: Master Specification v1.0.

Days in this log

- **2026-10-06:** P0 gate: power pull x10 passed after two real fixes (Wi-Fi lost at boot, confirmed writes lost); untrusted-certificate warning
- **2026-10-05:** P0 steps 1-9 built and deployed; gate: 3-device login, update + rollback, backup + restore, memory under load passed: hub base, schema + event log, access, devices, forgot PIN, client app; work log started
- **2026-10-03:** New network: own router, Pi re-flashed, SSH keys documented, HDMI fixed
- **2026-10-02:** P0 kickoff: spec review, dev PC tools, first Pi flash and first hub setup (TELUS network)

P0 gate: power pull ×10 passed after two real fixes (Wi-Fi lost at boot, confirmed writes lost); untrusted-certificate warning

Summary

- **New Android phone** ("1plus", Android 10, Chrome 151) worked without the certificate installed. Not a bug: Chrome lets a person tap through "Your connection is not private". The app now warns with an amber bar (DL-62).
- **Power-pull gate test (NFR-09, ×10): PASS on run 3.** 47,573 confirmed writes, **0 lost**, 0 duplicates, audit log complete, both databases intact, the real hub restarted by itself every time (48-78 s).
- It found two real problems first, both fixed and proven: 1. **Wi-Fi lost after power cuts during boot:** the Wi-Fi settings files were left empty. Fix: `chedam-netguard` (DL-60), tested by deleting the profile: recovered unattended in 127 s. 2. **Confirmed writes lost in a power cut** (23 of 1,646): SQLite confirmed before writing to the card. Fix: `synchronous=FULL` (DL-61).
- My test script crashed once (too short a timeout after a cold boot); fixed, and confirmed writes are now kept in a file as they happen.
- Gate status: 3-device login ✓, update + rollback ✓, backup + restore ✓, memory under load ✓, power pull ×10 ✓. Left: camera over HTTPS.

1. New phone works without the certificate

Hub record	"1plus", approved, Android 10, Chrome 151, app 0.9.0
Why it worked	Chrome's warning page "Your connection is not private" has Advanced → Proceed (unsafe) . After that the page works, but the hub is not verified (an impostor on the Wi-Fi would not be noticed) and Chrome refuses the app's offline mode (service worker) and may refuse the camera
Fix	The app detects the service worker's certificate error and shows an amber bar: "This device does not trust the hub's certificate yet, so offline mode and the camera will not work. Install the certificate " (link to the HTTP guide). Commit <code>61d86f0</code> , deployed

2. Power-pull test tools

File	Role
<code>tools/powertest/pi-powertest.sh</code>	<code>install</code> : a throwaway test hub that survives reboots (systemd service <code>chedam-powertest</code> , same binary, hooks, migrations and memory limits as the real hub, data in <code>/var/lib/chedam-powertest</code> on the same SD card, 127.0.0.1:8199); <code>check</code> : integrity of both databases, services, records vs. audit entries, power flag, boot id; <code>remove</code>
<code>tools/powertest/power-writer.mjs</code>	On the laptop through an SSH tunnel it restarts by itself: 3 writers (about 26 writes per second, each with its audit entry), notices the cut and the reboot (boot id), then checks: every confirmed write present, no duplicates, audit = records, integrity, real hub active

Dry run with a software reboot: 520 confirmed writes, 0 lost, PASS.

3. Run 1: the writer crashed, then the Pi came back without Wi-Fi

Time (UTC)	What happened
16:04	"Hub gone" for 6 s, no reboot: the laptop's SSH tunnel dropped (Wi-Fi blip); correctly not counted. Sreya had not pulled yet
16:06	Pull 1 → back after a reboot at 16:07

16:07	My script crashed while checking: listing ~17,000 records on a freshly booted Pi Zero took more than its 4 s timeout. The list of confirmed writes was only in memory, so this cycle could not be judged
after	Sreya pulled two more times; the Pi stopped answering (ping, SSH)

On the hub's monitor: a normal login prompt, but "My IP address is **127.0.1.1**" (no network address). The router listed 3 Wi-Fi devices, none the Pi. Sreya connected a keyboard:

Check	Result
<code>nmcli dev status</code>	wlan0 wifi disconnected
<code>rfkill list</code>	not blocked
<code>dmesg</code>	Wi-Fi firmware loaded normally (BCM43430/1)
<code>nmcli con show</code>	only 1o : no Wi-Fi network known at all

Sreya reconnected with `sudo nmcli dev wifi connect "TP-LINK_2CBE" password "..." ifname wlan0` (her password, typed by her), and the hub was back at 192.168.50.101.

Root cause

- Raspberry Pi OS 13 keeps the Imager Wi-Fi settings in `/etc/netplan/90-NM-*.yaml`, from which NetworkManager's profile is generated **at every boot**.
- Both files were **0 bytes**: they are rewritten during boot, and a power cut before the data reached the card left them empty (the new length was saved, the contents were not).
- Sreya's new profile carried yesterday's time (04:49), because the Pi's clock was wrong at boot: no network to set it and no RTC (the DL-23 risk, seen live).
- Data after these three cuts: both databases intact; 17,863 test records = 17,863 audit entries.
- My setup script only adds a power-save setting; it does not touch the Wi-Fi profiles.

Fix: chedam-netguard (DL-60)

- `hub/system/chedam-netguard + .service + .timer` (every minute), installed by `deploy.sh` and `setup-hub.sh`:
- **online**: an fsync'd spare copy of each valid Wi-Fi profile (must have an SSID and not be empty);
- **2 min without an address**: restore the spare copies if no Wi-Fi network is known, then reconnect;
- **10 min**: reboot, at most once an hour.
- **Controlled test on the Pi**: the profile was removed and Wi-Fi disconnected (with a 9-minute safety timer that would have put it back). Log:
- 09:43:55 profile removed;
- 09:46:33 "no Wi-Fi network known after 127 s ... restored spare copies";
- 09:46:37 "reconnected wlan0".

The hub was back at 192.168.50.101 with nobody touching it. The safety net was not needed (removed).

Fix: test script

- Checks after a reboot get 60 s with 3 attempts, and a failed check is reported, not a crash.
- Every confirmed write is appended to a file at once.

4. Run 2: confirmed writes lost (stopped after cycle 1)

Check	Result
Integrity, duplicates, audit, hub restart	all fine
Confirmed writes lost	23 of 1,646

- **Cause**: PocketBase opens SQLite in WAL mode with `synchronous=NORMAL`, which confirms a commit before forcing it onto the card. In a power cut the last confirmed transactions can vanish. That breaks NFR-02 ("no lost or

uplicated sales").

- **Fix (DL-61):** `hub/pb_hooks/durability.pb.js` sets `PRAGMA synchronous = FULL` on PocketBase's single non-concurrent (write) connection before every request, at start-up and every minute.
- Health page item "Saved before confirming" (bad if not FULL); test added; **329** tests pass.
- On the Pi: "Saved before confirming: Yes"; 30 writes in 2,147 ms (about 70 ms each, including the audit entry).
- Report kept: `docs/gate/powertest-2026-10-06-run2-synchronous-normal.json`.

5. Run 3: PASS 10/10

Cycle	Confirmed so far	Lost	Duplicates	Audit = records	Integrity real / test	Hub back after
1	873	0	0	yes	ok / ok	49 s
2	7,746	0	0	yes	ok / ok	48 s
3	10,239	0	0	yes	ok / ok	53 s
4	10,530	0	0	yes	ok / ok	51 s
5	10,775	0	0	yes	ok / ok	54 s
6	11,399	0	0	yes	ok / ok	64 s
7	11,770	0	0	yes	ok / ok	54 s
8	13,123	0	0	yes	ok / ok	65 s
9	13,845	0	0	yes	ok / ok	61 s
10	47,573	0	0	yes	ok / ok	78 s

- Each boot logged "EXT4 orphan cleanup", the normal filesystem repair after a power cut.
- Wi-Fi came back on its own every time (netguard did not need to act).
- Power was clean (`throttled=0x0`).
- 1-2 writes per cut were saved without being confirmed: they reached the card, but the answer was lost with the power. That is expected; the till retries them (idempotent ids in later phases).
- Report kept: `docs/gate/powertest-2026-10-06-run3.json`.
- The test hub was removed; the real hub is active.

6. Session 2: status check and a race fix

- Git: clean, `main` in sync with GitHub before starting. The Pi runs `dev-484c657`; the later commit `aebd3d5` was documents only.
- Full test run (`PB_BIN=C:/Tools/pocketbase.exe node hub/tests/run-all.mjs`): **1 failure**, `step9-updates` "one urgent task". It passed 3 times when run alone, so it was intermittent.
- **Cause (a real bug, not only the test):** when the hub applies an install result, it saved the update record and then its urgent task as two separate writes. The every-minute job and the status screen can both apply the same result at the same moment, so a reader could see "rolled back" before the task existed, and two appliers could each raise a task. `lib/backup.js` had the same pattern.
- **Fix (commit 85da79a):** in `hub/pb_hooks/lib/updates.js` and `hub/pb_hooks/lib/backup.js`, finding the record, saving it and raising/closing the task now run in one transaction. The hub has a single writer, so the second applier waits, then finds nothing left to apply.
- **Verified:** full run 329/329 passed; `step9-updates` and `step7-backups` then passed 5 times in a row each.
- Not deployed yet: needs Sreya's go-ahead.

7. Session 2 (continued): P0 closed, P1 started

P0 closed (DL-63)

- Sreya: skip the camera test; the certificate trust step can be solved later.
- Recorded as DL-63 in `docs/P0-mini-spec.md`: camera over HTTPS (R5) and the phone trust step move to P1

step 2 (phone scanning) and are tested before the P1 gate. The pilot week waits for a pilot store (Q9); P0 and P1 are piloted together.

P1 plan

- docs/P1-mini-spec.md: goal, 10 build steps with requirement IDs, decisions DL-64..71, open questions (Q1 accountant blocks the P1 release, not the build; Q2; Q7; who sees costs).

P1 step 1, hub side: catalogue and tax tables (commit d078aad)

File	What
hub/pb_migrations/1791300001_p1_catalogue_schema.js	tax_types, tax_rates (effective dates), tax_classes, deposits_fees, categories, products, selling_units (single, pack, case, by weight; nested), price_history
hub/pb_migrations/1791300002_p1_catalogue_reference.js	Permissions catalogue.edit (staff, manager), prices.edit and tax.manage (manager); GST 5%, BC PST 7%; classes Standard, GST only, Zero-rated, Exempt; all marked pending review (Q1)
hub/pb_migrations_dev/1791300100_dev_sample_catalogue.js	8 categories, 12 sample products (bananas by kg, cola single/12-pack/case, water single and 6-pack sharing a barcode, an age-restricted item, one Draft)
hub/pb_hooks/lib/catalogue.js	BR-07 checks (what keeps a product a Draft), base quantity of nested packs, barcode and PLU uniqueness, price history, the Drafts task, lookup and product view
hub/pb_hooks/lib/tax.js	Tax rates for a class at a date; checks on rates and classes
hub/pb_hooks/catalogue.pb.js	POST /api/chedam/catalogue/products (product + units in one transaction), GET /api/chedam/catalogue/lookup?code=, GET /api/chedam/catalogue/products/{id}
hub/pb_hooks/lib/event_log.js	Table rules run inside the save's transaction (SIDE_EFFECTS); "before" read from the database
hub/pb_hooks/lib/access.js	New tables; price changes and activation need prices.edit
hub/tests/step11-catalogue.test.mjs	67 tests

Command: PB_BIN=C:/Tools/pocketbase.exe node hub/tests/run-all.mjs → **396 passed, 0 failed** (11 suites).

Not deployed to the Pi yet (needs Sreya's go-ahead). Product screens are next.

Failed cases and fixes (session 2)

Symptom	Cause	Fix
Sample data migration failed: "base unit cannot change"	PocketBase's record.original() still shows the pre-create values when the same record object is saved a second time	Read the stored values from the database inside the transaction. The event log's "before" had the same weakness and now reads the database too
"Barcode 91 already belongs to Bananas"	In the JS hooks get() on a JSON field returns raw bytes (91 = "[")	Parse getString() (already a known gotcha)
A Draft showed outdated reasons	Reasons were computed when the product was created, before its units existed	Recomputed whenever a unit changes
400 answer lost the list of problems	PocketBase rewrites the data of thrown errors	The endpoint rolls back, then answers the list itself
13 tasks after loading sample data	Each product passed through Draft, so the Drafts task was created and closed 11 times	Reopen the last Drafts task (DL-71)

Migrations could not be reverted	Hard-deleting units re-validated their Active product	Hard deletes (superuser only) skip that refresh
Reversibility test missed the P1 migrations	It counted only names starting 17912	Counts every 1791... migration
Backups test failed sometimes ("already running")	The real every-minute backup job started a nightly backup during the test (any time after 02:30)	Test hubs switch that job off (CHEDAM_TEST_NO_BACKUP_CRON) and call the tick themselves; 5/5 passes
Review: nested packs, inner pack resized	A case did not follow when its inner pack changed size; nested packs of "each" items allowed fractions; an inner pack could be removed	Fixed, 3 tests added

Failed cases and fixes

Symptom	Cause	Fix
A device worked without the certificate	The person tapped through Chrome's warning	Amber warning bar in the app (DL-62)
"Hub gone" with no reboot	Laptop SSH tunnel dropped for a few seconds	Correctly ignored (boot id unchanged)
Writer crashed after pull 1	4 s timeout too short for a freshly booted Pi Zero listing ~17,000 records	60 s, 3 attempts; confirmed writes kept in a file
My journal query found nothing	The script logs in UTC, the Pi in Pacific time	Queried the right local time
Hub came back without Wi-Fi	Netplan Wi-Fi files emptied by a power cut during boot	chedam-netguard (DL-60)
\n turned into a real line break in a Python edit	Quoting layers	Fixed the line directly
23 confirmed writes lost	SQLite synchronous=NORMAL	synchronous=FULL (DL-61)
Intermittent test failure "one urgent task"	Update/backup result applied in two writes, by two callers at once	One transaction per result (85da79a)

Next

1. Sreya: go-ahead to deploy session 2's work to the Pi (race fix + P1 step 1 catalogue, with sample products).
2. P1 step 1 screens: product list, product form (units, packs, barcodes, tax class, Draft reasons), categories, tax tables.
3. P1 step 2: stock, add stock by phone, and the camera over HTTPS test (DL-63).
4. Open for Sreya: Q1 accountant review of the tax rules (blocks the P1 release), Q2 return defaults, who may see costs.

P0 steps 1-9 built and deployed; gate: 3-device login, update + rollback, backup + restore, memory under load passed: hub base, schema + event log, access, devices, forgot PIN, client app; work log started

Summary

- Full health check of the re-flashed Pi: healthy, OS up to date, no hub software yet.
- Restored admin access (the sudoers rule is lost on every re-flash).
- Changed `setup-hub.sh` so the display setting can be switched **both ways** (monitor on or headless) and re-runs leave the boot config unchanged.
- Ran hub setup with `--keep-display`, rebooted and verified everything. HTTPS works by name and by IP.
- iPhone 13 Pro and Android 9 installed the new "Chedam Store CA" and opened the hub page without warnings ("Chedam hub is running").
- Started this cumulative work log (`Work Log/`).
- P0 step 1 (hub base) is complete.
- P0 step 3 (access) is complete: permission engine with role templates, overrides and BR-33, PIN sign-in with lockout, owner recovery code, "Who can do this?". 92 more tests (124 in total) passing; deployed and checked on the Pi.
- P0 step 4 (devices) is built and tested on the PC (67 new tests, 192 in total): pairing by code/QR or by approval request, device key checked on every request, name-then-PIN sign-in on paired devices only, one person per device, device manager (approve, lock, sign out, rename, assign, remove), certificate guide. Deployed to the Pi and checked; pairing the real phones is next.
- Forgot PIN decided and built: a manager sets a temporary PIN, and the person chooses their own at the next sign-in.
- P0 step 5 (client shell) is built and tested on the PC: an installable Svelte app that opens offline, with the connectivity bar, name-then-PIN sign-in, choosing a new PIN, owner sign-in, idle auto-lock, large text and high contrast, and the device manager with QR codes. It replaces the temporary pages. 213 hub tests pass.
- P0 step 6 (setup wizard) is built, tested and deployed: one-time setup code on a new hub, owner and first device in one step, recovery code, business profile, logo and receipt, team, storage, features, references, backups; skipped steps become tasks. 251 hub tests pass.
- P0 step 7 (backups) is built and tested **on the Pi** with a disk image: encrypted, verified backups, restore test, and a real restore in 2-3 s that keeps the HTTPS certificate authority. Sreya's hard drive showed read errors, so nothing was written to it. 281 hub tests pass.
- **3 devices signed in as the owner** (iPhone, Android 9 phone, Surface laptop) after fixing a sign-in mix-up and the laptop certificate. Issue logged: the Android 9 phone runs Chrome 98, too old for the app's styles (Tailwind 4 needs Chrome 111+), so its screens are misaligned.
- P0 step 8 (hub health page) is built and deployed; 302 hub tests pass.
- P0 step 9 (updates) is built and tested on the Pi: a signed install in 1.7 s, and a broken update rolled back automatically in 4 s; tampered packages are refused. 323 hub tests pass.
- First backup to the new SD card made by Sreya and verified independently; **0.9.0 published** to `chedam-releases` and downloaded and verified by the Pi from GitHub.
- P0 gate, memory under load: 10 devices for 5 minutes peak at **108 MB** (budget 150 MB). A stress burst first reached 194 MB; per-device rate limits now hold it at 120 MB.
- Re-tested the step 3 bugs and issues: all fixed. A third copy of one bug (recovery code) was found and fixed.
- P0 step 2 (schema v1 + event log) is complete: 14 tables, reference data, dev sample store, a transactional append-only event log, 32 automated tests passing, deployed to the Pi with a backup first.

Naming note: the local folder `chedam-updates/` in the project is Sreya's YouTube and notes repo (gitignored). It is **not** the update channel. Nothing is renamed; the public update-package repo name will be decided when step 9 (Updates) starts.

1. Health check (before any change)

One SSH session ran these checks:

```
whoami; hostname; hostname -I; cat /proc/device-tree/model; grep PRETTY /etc/os-release; uname -r
uptime; sudo -n true; free -m; df -h / /boot/firmware; lsblk
vcgencmd measure_temp; vcgencmd get_throttled; vcgencmd measure_volts core
timedatectl; ip route; ping -c2 1.1.1.1
apt list --upgradable; dpkg-query -W <packages>; systemctl is-active ...; systemctl --failed
cat /boot/firmware/config.txt /boot/firmware/cmdline.txt
```

Area	Result
Hardware	Pi Zero 2 W Rev 1.0, 33.2 °C, throttled 0x0, core 1.2875 V
OS	Debian 13 trixie, kernel 6.18.50+rpt-rpi-v8, 0 pending updates
Uptime	1 day 12 h, load 0.00, no failed units
Memory / disk	415 MB total, 273 MB available / 119 GB card, 2.3 GB used
Network	192.168.50.101 via 192.168.50.1, internet ping 7 ms
Time	NTP synced, America/Vancouver, no RTC
sudo	Needed a password (re-flash reset it)
Hub software	Not installed: no caddy, chrony, ufw, sqlite3 or <code>/opt/chedam</code>
Display	<code>video=HDMI-A-1:1280x720@60D</code> present (monitor in use)

2. Admin access

Failed case: `Permission denied (publickey)` when restoring sudo

- The `ssh -t chedam "...sudo tee..."` command was typed **inside an SSH session on the Pi** (prompt `chedam@chedam:~ $`).
- So the Pi tried to SSH into itself and had no key for that. Re-flashing was not needed.
- Fix: run only the inner command on the Pi:

```
echo 'chedam ALL=(ALL) NOPASSWD:ALL' | sudo tee /etc/sudoers.d/010-chedam && sudo chmod 440 /etc/sudo
sudo -n true && echo OK
```

3. Display becomes a switch (DL-24 updated)

Sreya's requirement: the monitor is optional on the Zero 2 W pilot, but the option must stay. Pi 4 production hubs will use a display to operate the hub directly.

Change in `hub/scripts/setup-hub.sh` (boot config section):

- Removes any earlier chedam block, then writes a block between `# chedam-begin` and `# chedam-end`.
- `--keep-display`: display driver `vc4-kms-v3d` and display detection **on**, Bluetooth off.
- No flag: headless. The display driver and detection are off, `gpu_mem=16`, Bluetooth off; this saves ~50 MB of RAM.
- Audio and camera detection are always off.

```
sudo bash ~/hub/scripts/setup-hub.sh --keep-display # monitor on (current)
sudo bash ~/hub/scripts/setup-hub.sh                # headless
sudo reboot                                           # after switching
```

Failed case: an extra `[a11]` line on every run

- The test (switching off, on, off, on on a copy of `config.txt`) worked, but each run appended another `[all]` line. It was harmless but untidy.
- Fix: moved `[all]` inside the marked block. The original config (`config.txt.chedam-orig`, saved by the first run) was restored and setup run twice. Both runs gave the same checksum (`d91ff929`), and the only differences from the original were audio off, camera detection off and the chedam block.

4. Hub setup on 192.168.50.101

```
scp -r scripts system chedam:hub/
# on the Pi: download + verify PocketBase
wget pocketbase_0.40.4_linux_arm64.zip checksums.txt
grep linux_arm64 checksums.txt | sha256sum -c -      # OK
sudo apt-get update
sudo bash ~/hub/scripts/setup-hub.sh --keep-display
sudo systemctl reboot
```

Verification after reboot:

Check	Result
Services	chedam-hub, caddy, chrony, avahi-daemon, ufw active; no failed units
Memory	PocketBase 28 MB + Caddy 37 MB = 65 MB (budget 150 MB, NFR-08); 264 MB available with display on
Watchdog	RuntimeWatchdogUsec=14s
Time	chrony, 0.0000007 s from NTP, leap normal
Temperature	38.6 °C, throttled 0x0
Display	/dev/dri/card0 present
Firewall	22, 80, 443/tcp and 5353/udp allowed (IPv4 + IPv6)
CA download	<code>http://chedam.local/ca.crt</code> → 200, "Chedam Store CA - 2026 ECC Root", valid to 2036-08-13
HTTPS	<code>https://chedam.local/api/health</code> and <code>https://192.168.50.101/api/health</code> → 200
Certificate check	<code>openssl s_client ... -CAfile ca.crt</code> → Verify return code 0, with and without SNI
HTTP	Redirects to HTTPS (302)
Caddy errors since boot	0

PC test commands (Git Bash; `--resolve` avoids Windows mDNS problems, and `--ssl-no-revoke` avoids the Schannel revocation error):

```
curl --resolve chedam.local:80:192.168.50.101 -o ca.crt http://chedam.local/ca.crt
curl --ssl-no-revoke --resolve chedam.local:443:192.168.50.101 --cacert ca.crt https://chedam.local/a
curl --ssl-no-revoke --cacert ca.crt https://192.168.50.101/api/health
```

5. Phones (risk R5, first part)

Device	How	Result
iPhone 13 Pro, iOS 26.6.1	Safari <code>http://chedam.local/ca.crt</code> → install profile → Settings > General > About > Certificate Trust Settings → on	<code>https://chedam.local</code> shows the placeholder, no warning
Android 9	Chrome <code>http://192.168.50.101/ca.crt</code> → install for VPN and apps	<code>https://192.168.50.101</code> shows the placeholder, no warning

The camera test over HTTPS comes later, when the client app exists.

6. Work log started

- Folder `Work Log/`: one Markdown source per day in `Work Log/entries/YYYY-MM-DD.md`.
- `tools/worklog/build_worklog.py` builds `Work Log/Chedam-Work-Log-YYYY-MM-DD.pdf` with **all** days, newest first.
- Earlier PDFs are kept, so each day has its own complete snapshot.

```
python tools/worklog/build_worklog.py          # newest day
python tools/worklog/build_worklog.py 2026-10-05 # a specific day
```

The days 2026-10-02 and 2026-10-03 were written up from the session records and `docs/P0-pi-setup-log-2026-10-03.pdf`.

7. P0 step 2: schema v1 and event log

Spec basis: Section 10 (data model, event log), Section 13 (build process: migrations and sample data first), NFR-20 (reversible migrations), BR-30 (UTC), BR-33 (managers cannot grant more than they hold).

Files added

File	Purpose
hub/pb_migrations/1791200001_p0_schema_v1.js	13 new tables + the PocketBase <code>users</code> collection extended: business, locations, settings, modules, permissions, roles, permission overrides, devices, storage_areas, tasks, events, backups, updates. Common fields on every table. API rules all closed (superuser only) until step 3
hub/pb_migrations/1791200002_p0_reference_data.js	22 modules (3 core always on, 15 switchable, 4 later), shop-type presets, 20 P0 permissions (5 owner-only), 5 role templates, 12 default settings
hub/pb_migrations_dev/1791200100_dev_sample_store.js	Dev-only sample store: "Demo Grocery", main location, grocery preset modules, 4 storage areas, 5 people (one per role, random unknown passwords), 1 setup task
hub/pb_hooks/event_log.pb.js + lib/event_log.js	Event log: one <code>events</code> row per create/update/delete, in the same transaction; actor and device stamped from the request; append-only
hub/tests/step2-schema-eventlog.test.mjs	32 automated tests against a throwaway local PocketBase
hub/scripts/deploy.sh	PC → hub deploy: copy, SQLite online backup + integrity check, rsync, restart, health, migrations, errors
docs/conventions.md	Data, event-log, migration and workflow rules for all phases

Design choices

- **Common fields:** `id` (a device may create it offline), `created_at`, `updated_at`, `created_by`, `updated_by`, `device_id`, `deleted_at`. `updated_by` was added beyond the spec's list, so each change records who made it.
- **The event log runs inside the same transaction** as the change (`onRecord*Execute + runInTransaction`). If the event can't be written, the change is rolled back, so the log never misses a write.
- **Actor and device are stamped by the hub** in request hooks. A client cannot forge `created_by`, and the device comes from the `X-Chedam-Device` header (verified in step 4).
- **Hidden fields** (`pin_hash`, `recovery_code_hash`, `key_hash`, `pairing_code_hash`) never reach the API or the log.
- **Role levels:** owner 100, manager 80, accountant 50, cashier 40, staff 30. Owner-only permissions: `modules.manage`, `setup.run`, `owner.manage`, `backups.restore`, `updates.install`.
- **Sample data is separate** from real migrations, so a store hub never receives demo records.

Commands

```
# local tests (PC, Git Bash)
PB_BIN='C:\Tools\pocketbase.exe' node hub/tests/step2-schema-eventlog.test.mjs
# deploy to the Pi with sample data
bash hub/scripts/deploy.sh --sample-data
# verify on the Pi
ssh chedam 'sudo -u chedam-hub sqlite3 /opt/chedam/pb_data/data.db "select table_name, action, count('
```

Failed cases and fixes

Symptom	Cause	Fix
pocketbase.exe: command not found in Git Bash	That shell started before C:\Tools was added to PATH	Full path C:\Tools\pocketbase.exe / PB_BIN
migrate down hung	It waits for a y/N confirmation	Pipe y (input: "y\n" in the tests)
Python check query error (ESCAPE expression must be a single character)	Backslash escaping in an inline script	Filter with <code>substr(name,1,1) != '_'</code>
Sample migration: Cannot read property 'forEach' of undefined	A JSON field read with <code>get()</code> returns raw bytes	<code>JSON.parse(record.getString("value"))</code>
Test stop step hung	Waited for the exit of a process that had already exited	Check <code>exitCode/signalCode</code> first
Create returned 400 "Something went wrong"	<code>e.request</code> doesn't exist in PocketBase 0.40 JS hooks (found with <code>serve --dev logs</code>)	<code>e.requestInfo().headers["x_chedam_device"]</code> (names lowercased, <code>_</code> → <code>_</code>)
A Python edit put real newlines inside JS strings	Escaping in a heredoc	Fixed with a direct edit

Test results (PC, 32/32 passed)

- Schema/reference: 22 modules, 3 core on, 15 switchable, 20 permissions, 5 roles, owner holds all, manager holds no owner-only permission, 12 settings.
- Sample: business + location, 4 storage areas, 5 people, 1 task, 7 grocery modules on.
- Event log: device-made id accepted; forged `created_by` overwritten; device stamped; create/update/soft-delete/delete each logged with correct before/after/changed; delete keeps the last state; no PIN hash, password or tokenKey in the log; events can't be edited or deleted, even by a superuser.
- Atomicity: with the event write forced to fail, the create fails and the record is not in the table.
- Migrations: down 3 and up again both clean.

On the Pi (after deploy)

Check	Result
Pre-deploy backup	<code>pre-deploy-<UTC>.db</code> , integrity ok
Migrations applied	schema v1, reference data, dev sample store
Rows	modules 22, permissions 20, roles 5, settings 12, business 1, locations 1, storage_areas 4, users 5, tasks 1
Events	78 = every write above (incl. 7 module updates for the grocery preset)
Secrets in events	0
SQLite journal	WAL
PocketBase memory	42 MB (budget 150 MB)
Errors in the journal	none

8. P0 step 3: Access

Spec basis: FR-1.03 (owner password, PIN, printed recovery code), FR-1.04 (people with role and PIN), FR-1.09 (owner signs in on any device), FR-1.10 (role templates, per-person overrides with end date, "Who can do this?"), NFR-11 (hub-side permission checks, PIN lockout after 5 tries, auto-lock), BR-33 (managers can't grant more than they hold).

Files added or changed

File	Purpose
------	---------

hub/pb_migrations/1791200003_p0_access.js	PIN and recovery code become bcrypt password fields (hidden); pin_set and recovery_code_created_at added; API rules set on all 14 tables; token lifetime 12 h. Reversible
hub/pb_hooks/lib/access.js	Permission engine: effective permissions (role + allow overrides – deny overrides; owner = all), the per-table access map, protected fields, self/assignee field limits, BR-33 checks for people, roles, overrides, modules, settings, business
hub/pb_hooks/lib/auth.js	PIN rules, lockout counter, recovery code generator, secret check
hub/pb_hooks/access.pb.js	Guard on every list/view/create/update/delete; password sign-in lockout; token refresh check; endpoints (below)
hub/pb_migrations_dev/1791200101_dev_sample_pins.js	Dev-only PINs for the 5 sample people (test values; dev hubs only)
hub/pb_hooks/lib/event_log.js	Fix: put the original app back after the transaction (see failed cases)
hub/tests/lib/hub.mjs, step3-access.test.mjs, run-all.mjs	Shared test harness, 92 access tests, one-command runner
docs/conventions.md, docs/P0-mini-spec.md	Access rules for later phases; decisions DL-28 to DL-32

Endpoints

Method and path	Who	What
GET /api/chedam/auth/pin-users	anyone on the LAN (step 4: paired devices only)	Names for the till sign-in screen
POST /api/chedam/auth/pin	anyone	{user, pin} → sign-in token; wrong tries count toward lockout
POST /api/chedam/users/{id}/pin	yourself, or users.manage above them	Set or change a PIN (4-6 digits, no repeats or straight runs)
POST /api/chedam/users/{id}/unlock	users.manage above them	Clear a lockout
POST /api/chedam/owner/recovery-code	owner	New printable code, shown once (format xxxxx-xxxxx-xxxxx-xxxxx)
POST /api/chedam/auth/recover	anyone with the code	{email, code, new_password} → new password + sign-in; the code is used up
GET /api/chedam/access/me	signed in	My role and effective permissions
GET /api/chedam/access/users/{id}	self or users.view	Someone's effective permissions and where each comes from
GET /api/chedam/access/who-can/{code}	users.view	"Who can do this?" (via role or override, with end date)

Who may do what (summary of the access map)

- **Anyone signed in:** reads the business profile, locations, settings, modules, permissions, roles and storage areas.
- **People:** list with users.view; a person can view and edit their own record (name, email, phone, language, accessibility, password only); add/edit others needs users.manage; giving or changing roles needs access.manage; no hard delete.
- **Overrides:** access.manage only; never on yourself; only on people below you; never grant what you don't hold; owner-only permissions only by the owner.
- **Roles:** templates can't be deleted; only the owner edits templates; the owner role always holds everything; managers create and edit only roles below their own level.
- **Modules:** owner only; only enabled can change; core can't be switched off; "later" modules can't be switched on; who/when stamped.

- **Settings:** `security.*` owner only; keys can't be renamed.
- **Audit log:** `events.view`. Tasks: `tasks.view` / `tasks.manage` (the assignee may update status/note). Devices: `devices.view` / `devices.manage`. Backups and updates: read-only through the API.

Failed cases and fixes

Symptom	Cause	Fix
No way to check a bcrypt PIN in JS hooks	<code>\$security</code> has no bcrypt	Use a PocketBase <code>password</code> field: <code>record.getRaw("pin").validate(plain)</code>
Owner recovery sign-in failed: failed to delete all previous MFAs ... transaction has already been committed	The event-log wrapper replaced <code>e.app</code> with the transaction and never restored it, so PocketBase's clean-up after a password change used a finished transaction. This would have broken every password change.	Restore <code>e.app</code> in a <code>finally</code> block after <code>runInTransaction</code>
5th wrong password returned 400, not 423	The failure was counted, but that same response didn't report the new lock	After counting, re-read the person and answer "locked" (same as PIN sign-in)
Assumed PocketBase leaves <code>e.record</code> empty on a wrong password	Traced with debug lines: the hook gets the person and <code>e.next()</code> checks the password	Removed the extra lookup and the debug lines
Step 2 test still sent <code>pin_hash</code> and reversed only 3 migrations	The schema changed in step 3	Test uses the <code>pin</code> field, checks for no <code>\$2a\$</code> hashes, and reverses every Chedam migration it finds
<code>sed</code> edits to the test harness didn't apply	Escaping in shell quoting	Direct edits

Test results (PC)

`node hub/tests/run-all.mjs` → step2 32/32, step3 92/92.

Step 3 covers: - PIN sign-in, the PIN list showing no secrets, and wrong PINs refused. - Lockout after 5 wrong PINs and after 5 wrong passwords (423), and unlock by a manager but not a cashier. - The PIN rules. - Who can read and change each table. - Managers adding cashiers but not owners or other managers, and editing people below them but not the owner. - Role changes, and the rule that you can't change your own role. - Suspending someone signs them out at once. - Overrides: allow, deny, past and future end dates, owner-only, and "can't grant what you don't hold". - Role templates, the owner role, and deleting custom roles. - The module, setting and business rules. - The recovery code: shown once, case-insensitive, used once. - Who-can and my-permissions. - The audit trail: lockouts, overrides and module switches are logged, and no hashes appear anywhere.

On the Pi (after deploy, checked on the Pi itself)

Check	Result
Deploy	Backup + integrity ok; migrations <code>1791200003_p0_access.js</code> and <code>1791200101_dev_sample_pins.js</code> applied; no errors
PIN list	5 people
Signed-out request	401
PIN sign-in time	0.50 s (bcrypt cost 10 on the Pi Zero 2 W)
<code>access/me</code> for the cashier	role cashier, permissions <code>tasks.view</code>
Cashier lists people	403
bcrypt hashes in the event log	0
PocketBase memory	44 MB

The dev PINs are in the dev seed file only and are not repeated in this log.

9. Re-test of the step 3 bugs and issues

Sreya asked which 2 bugs and 2 issues came up in step 3, whether they were fixed, and to test them again.

#	Kind	What	Status after re-test
1	Bug	Event-log wrapper left <code>e.app</code> pointing at a finished transaction, so the owner's recovery sign-in (and every password change) failed	Fixed. Tests "recovery code + new password signs the owner in" and "new password works" pass
2	Bug	The 5th wrong password answered 400 instead of 423 (locked)	Fixed. Test "5 wrong passwords lock the owner too" passes
3	Issue	PocketBase JS hooks have no <code>bcrypt</code> function	Solved with PocketBase <code>password</code> fields; PIN tests pass
4	Issue	The step 2 test was out of date after the step 3 schema change	Updated; step 2 suite passes (now 33 tests)
5	Bug found now	Same as bug 2 on the recovery code path: the 5th wrong code answered 400, not 423	Fixed: one shared <code>auth.failAndThrow()</code> now serves PIN, password and recovery code. New test "5th wrong recovery code answers 423"

Baseline before any change: `node hub/tests/run-all.mjs` → 32/32 and 92/92.

10. P0 step 4: Devices (FR-1.07, 1.08)

Files added or changed

File	Purpose
<code>hub/pb_migrations/1791200004_p0_devices.js</code>	<code>paired_via</code> , <code>paired_at</code> on devices; generic create closed; settings <code>devices.pairing_code_minutes</code> (10), <code>devices.online_seconds</code> (120), <code>devices.max_pending_requests</code> (10). Reversible
<code>hub/pb_hooks/lib/devices.js</code>	Device check for every <code>/api/</code> request, pairing codes, keys, last seen, device view
<code>hub/pb_hooks/devices.pb.js</code>	Middleware + endpoints (below)
<code>hub/pb_hooks/access.pb.js</code>	Password and PIN sign-in need a paired device (owner excepted); sign-in records who is on the device; assigned devices; recovery-code fix
<code>hub/pb_hooks/lib/access.js</code>	Devices: generic create closed, protected fields, BR-33 on assigning
<code>hub/pb_hooks/lib/auth.js</code> , <code>lib/event_log.js</code>	<code>failAndThrow()</code> ; device stamped only when verified
<code>hub/pb_public/</code>	Temporary pages until the Svelte client (step 5): <code>device-setup.html</code> (certificate guide, works over HTTP), <code>pair.html</code> (pair, then pick your name and enter PIN), <code>devices.html</code> (device manager with QR code), <code>assets/</code> (shared CSS/JS, vendored QR library, <code>VENDORED.md</code>)
<code>hub/system/Caddyfile</code>	HTTP serves the guide and its CSS; <code>/hub.json</code> gives the hub IP for QR codes
<code>hub/scripts/deploy.sh</code>	Also updates the Caddy config when it changed (keeps the IP, validates, keeps a copy, reloads)
<code>hub/tests/lib/hub.mjs</code> , <code>step2...</code> , <code>step3...</code> , <code>step4-devices.test.mjs</code>	<code>pair()</code> helper and automatic device headers; existing suites now sign in on paired devices; 67 new tests
<code>docs/P0-mini-spec.md</code> , <code>docs/conventions.md</code>	DL-33 to DL-38, device rules, the open "forgot PIN" item

Endpoints

Method and path	Who	What
POST <code>/api/chedam/devices/pairing-code</code>	devices.manage	<code>{name, type}</code> → one-time code <code>xxxx-xxxx</code> (10 min)
POST <code>/api/chedam/devices/pair</code>	new device	<code>{code}</code> → device id + key, approved
POST <code>/api/chedam/devices/request</code>	new device	<code>{name, type}</code> → id + key, waits for approval
GET <code>/api/chedam/devices/me</code>	the device	Its status and who is signed in (works while pending or locked)

POST /api/chedam/devices/me/sign-out	the device	Sign out whoever is on it
GET /api/chedam/devices	devices.view	Device manager list: online, signed in, assigned, version, last seen, browser
POST /api/chedam/devices/{id}/approve, lock, unlock, sign-out, revoke	devices.manage	Device manager actions
PATCH /api/collections/devices/records/{id}	devices.manage	Rename, type, assign person or printer

Sign-in flow (Sreya: "select name and then PIN")

The device is paired once. After that the sign-in page shows the people's names as large buttons. Tap your name, enter your PIN on the keypad and press OK. A wrong PIN shakes the keypad and shows "Wrong name or PIN". After 5 wrong tries the person is locked (423). "Not you? Back to names" returns to the list. The owner also has "sign in with email and password", which works on any device. "Forgot your PIN?" shows the current options (see the open item in the mini-spec).

Commands

```
PB_BIN='C:\Tools\pocketbase.exe' node hub/tests/run-all.mjs
npm pack qrcode-generator@2.0.4 # vendored QR library; integrity checked against the npm register
```

Failed cases and fixes

Symptom	Cause	Fix
Shell edit script stopped with "unexpected EOF while looking for matching quote"	Quotes and backticks of JS code inside a bash heredoc	Wrote the Python edit script to a file and ran it
Helpers at the top of devices.pb.js would be undefined in handlers	Each PocketBase JS handler runs in its own VM and cannot see top-level variables of the .pb.js file	Moved them into lib/devices.js (loaded with require inside each handler)
Test "key and code hashes never in API output" failed	The test read the record as superuser , and PocketBase shows hidden fields to superusers on purpose	The test now reads as the manager (who never sees them); superuser = development admin only
"Paired as ..." message vanished at once	The status refresh after pairing cleared the message	Show the message after the refresh
PIN typed by the browser test tool did nothing	The tool inserts text without key events (a real keyboard sends them)	Tested by tapping the on-screen keypad, as on a phone
Deploy to the Pi blocked	Deploying to the hub needs Sreya's approval in this session	Sreya approved; deployed

Test results (PC)

node hub/tests/run-all.mjs → step2 33/33, step3 92/92, step4 67/67 (192 in total).

Step 4 covers: - **Pairing codes**: who may make them, the format, lower case with a space accepted, single use, expiry, approved by the code's maker. - **Device keys**: unknown, wrong or missing keys refused (401); keys never in API output or the log. - **Ask to join**: pending until approved; the cap on open requests (429). - **Device needed**: no PIN list or PIN sign-in without a device; a cashier's password sign-in or token without a device refused; the owner works on an unpaired browser. - **One person per device**: the earlier token is refused, tokens can't move devices, sign-out. - **Device manager**: list (online, signed in, version, own device marked); sign-out, lock (423)/unlock, rename; protected fields; no generic create. - **Assigned devices**: list and sign-in limited; BR-33 on assign and remove. - **Revoke**: device and token refused, no double remove. - **Recovery**: the 423 fix, and owner recovery on a till. - **Log**: last-seen writes throttled (≤1 event for 10 requests), lock/unlock/revoke logged with actor and the manager's device, no hashes. - **Guessing**: 20 wrong codes pause pairing (429).

Browser check (local dev hub, PC)

Paired with a code from the URL. The name list showed 5 sample people. A wrong PIN gave "Wrong name or PIN" and the right PIN signed in (manager), with the Device manager link shown. The device manager showed the till online with

its version and browser. Pair a new device showed the QR code, the code and a countdown. At phone width (375 px) the certificate guide opened the Android steps automatically, with no sideways scrolling.

On the Pi (after deploy, checked from the PC over the LAN)

```
bash hub/scripts/deploy.sh --sample-data
```

Check	Result
Deploy	Backup integrity ok; 1791200004_p0_devices.js applied; Caddy config updated for 192.168.50.101 and reloaded (old copy in backups/); no errors
Certificate guide over plain HTTP	http://192.168.50.101/device-setup.html 200, its CSS 200; other HTTP pages still redirect to HTTPS (302)
https://192.168.50.101/hub.json	{"ip":"192.168.50.101","name":"chedam.local"}
Pages over HTTPS	pair.html, devices.html, assets/qrcode.js 200
PIN list without a device	403
Made-up device header	401
Wrong pairing code	400
Device settings	3 devices.* settings present
PocketBase memory	40 MB (budget 150 MB)
Temperature	37.0 °C

11. Forgot PIN: temporary PIN (Sreya's decision)

Sreya: "manager reset temporary and user need to reset again".

- A PIN set by someone else (manager with users.manage above the person, or superuser) is **temporary**. It works for 24 h (security.temp_pin_hours).
- With a temporary PIN, the hub allows only these until a new PIN is chosen: setting your own PIN, access/me, devices/me and sign-out. Everything else answers 403 "Choose your own new PIN first." (auth.pinChangeGate, run in the request middleware).
- The new PIN must differ from the temporary one, and the PIN rules apply. An expired temporary PIN is refused with a clear message.
- pin_must_change and pin_temp_expires_at are protected fields, and both changes are in the audit log.
- Migration 1791200005_p0_temp_pin.js (reversible). Tests: step3b-temp-pin.test.mjs, 14/14.
- In the app: the manager uses "Someone forgot their PIN" on the home screen; the person then sees "Choose your own PIN" (enter twice).

12. P0 step 5: client shell (FR-1.09, FR-12.01, NFR-16/17)

Hub side

File	Purpose
hub/ pb_migrations/1791200006_p0_connectivity.js	Settings network.check_url (Cloudflare's 204 check) and network.hotspot_ssids
hub/pb_hooks/lib/net.js, status.pb.js	Check once a minute (cron); GET /api/chedam/status → {internet: offline/router/hotspot, hub_time}; open to waiting/locked devices too
hub/pb_hooks/access.pb.js	access/me also returns pin_must_change, large text, high contrast, language
hub/tests/step5-status.test.mjs	7 tests: open, values, hub clock, offline/online via the check address, guests can't force a check, waiting device

Client (client/)

Part	What
Stack	Svelte 5.57.1, Tailwind 4.3.3, Vite 8.3.2, qrcode-generator 2.0.4 (same integrity hash as checked in step 4). Exact versions; <code>npm audit</code> 0 vulnerabilities
Build	<code>npm run build</code> → <code>hub/pb_public/</code> (about 35 KB gzipped JS + 4 KB CSS). <code>deploy.sh</code> builds it before copying
PWA	Manifest, icons (drawn by <code>client/scripts/make-icons.mjs</code> , no image library needed), service worker generated with the exact file list
Screens	Pair (code / ask to join), Waiting (approval or locked), Names, PIN pad, Choose your own PIN, Owner sign-in, Home (modules switched on, "Someone forgot their PIN", My settings, Install), Devices (manager)
Connectivity bar	Online (router) / Online (hotspot) / Offline, "Hub not reachable", device clock warning (2+ min)
Auto-lock	After <code>security.auto_lock_minutes</code> idle: signed out on that device, name list returns
Removed	The temporary step 4 pages (<code>pair.html</code> , <code>devices.html</code> , <code>assets/chedam.js</code> , vendored <code>qrcode.js</code>). The certificate guide moved to <code>client/public/</code> and now opens the app with <code>/?pair=CODE</code>
Caddy	HTTP serves <code>/device-setup.html</code> and <code>/setup.css</code>

Browser check (local dev hub on the PC)

Check	Result
Session kept	The earlier paired till and Mira's session came back after the app update
Connectivity bar	"Online (router)"
Forgot PIN, end to end	Mira set a temporary PIN for Sam → Sam signed in with it → forced to "Choose your own PIN". A mismatch was caught, reusing the temporary PIN was refused, the new PIN was set and Sam reached his home screen (no Devices tile: staff)
Offline shell	Service worker active, 8 files cached. With the hub stopped , a reload still opened the app with the red "Hub not reachable" bar and a clear message
Recovery	Hub started again → bar back to "Online (router)"; Sam's session came back
Device manager	Mira → Devices → Pair a new device → QR code + code + countdown
Phone size, dark mode	Readable, no sideways scrolling
Certificate guide	Page and <code>setup.css</code> 200; continues to <code>/?pair=CODE</code>

Failed cases and fixes

Symptom	Cause	Fix
Step 2 test "12 default settings" failed (13)	It counted settings; step 4/5 add their own	Checks the 12 step-2 keys by name
Svelte warning: <code>pad</code> not declared with <code>\$state</code>	Component reference in a plain variable	<code>let pad = \$state()</code>
Offline reload showed a blank page	The module script is requested with an <code>Origin</code> header and the hub answers <code>Vary: Origin</code> , so the cached copy (stored without <code>Origin</code>) did not match	Cache look-ups use <code>ignoreVary: true</code> (our files are content-hashed)
Crash "Cannot read properties of null (reading 'user')" when signing out	Home still drew <code>s.me.user</code> for a moment after sign-out cleared it	Home renders only while someone is signed in
After the hub came back, the name list kept its error until the next 30 s poll	Nothing reacted to the hub returning	The app refreshes as soon as the hub answers again; the name list retries
First page load after restarting the dev hub was refused by the browser pane	The page was opened before the server was listening	Waited 2 s and opened it again

Test results (PC)

node hub/tests/run-all.mjs → step2 33/33, step3 92/92, step3b 14/14, step4 67/67, step5 7/7 (213).

On the Pi (after deploy, Sreya's go-ahead)

Check	Result
Deploy	App built (35 KB gzipped JS); backup integrity ok; migrations ...05_p0_temp_pin, ...06_p0_connectivity applied; Caddy updated and reloaded; no errors
App over HTTPS (IP and chedam.local)	/, sw.js, manifest.webmanifest, icons: 200
Certificate guide over HTTP	device-setup.html, setup.css: 200
/api/chedam/status	internet: router; hub clock matches
Wi-Fi name readable by the hub (nmcli)	yes, so hotspot detection works
PocketBase memory	40 MB

13. Phones paired; bug: the app froze on the iPhone with the Pi off

Sreya paired the Android 9 phone ("9lite") and the iPhone; both appear in the device list. With the Pi powered off, the app on the iPhone froze: no button did anything.

Cause	When the hub is switched off, a phone's requests are not refused; they wait (nothing answers at that address). (1) The service worker waited for page loads with no time limit . (2) Each button waited up to 8 s for the hub, some twice in a row, with no feedback. (3) The status bar checked only every 30 s. The PC test had missed it: a stopped server on the same machine refuses at once
Fix	Page loads get 3 s, then the cached app opens. Requests get 6 s; once the hub is known to be down, requests fail at once until it answers again. The bar checks every 5 s while the hub is down. Signing out with the hub off returns to the name list. "Check" shows "Checking..."
Test	A test server that accepts connections and never answers (like a switched-off hub seen from a phone): bar red after 4.0 s; sign-out 51 ms; reload shows the app at once with the red bar; real hub started again (page not touched): back to "Online (router)" with the name list after 4.3 s
Deployed	Commit 6ede814; deploy ok (backup integrity ok, no errors)

Note: a phone gets the fixed version the next time it opens the app **while the Pi is on**. Open it, wait a few seconds, then close and reopen once.

14. P0 step 6: setup wizard (FR-1.01-1.06, 1.13, 1.15)

Sreya chose: **a setup code on the hub** for the first device (DL-44).

Hub

File	Purpose
hub/ pb_migrations/1791200007_p0_setup.js	devices.paired_via gains "setup" (reversible)
hub/pb_hooks/lib/setup.js, setup.pb.js	Setup code (made at startup, by a once-a-minute job and on demand; kept in pb_data/setup-code); monitor message pb_data/console.issue: GET /api/chedam/setup/status, POST /api/chedam/setup/start, POST /api/chedam/setup/step
hub/pb_hooks/lib/access.js	business.setup_state changes only through the step endpoint
hub/system/chedam-console.path + .service	When the hub's message changes, the monitor's login screen re-reads it (agetty --reload); /etc/issue.d/chedam.issue links to the message
hub/scripts/setup-hub.sh, deploy.sh	Install the monitor units and link; print the setup code on a new hub
hub/tests/step6-setup.test.mjs	38 tests on a hub without sample data

The 38 tests cover: - **The code**: made on a new hub, kept across restarts, shown in the monitor message. - **Refused at start**: a wrong code, a short password, a guessable PIN and a bad email. - **Start**: returns the device key and recovery

code once. Afterwards the code file is gone, the monitor message changes and the code can't be reused. - **The new owner:** PIN, password and recovery code all work. The first device is approved and the owner is signed in on it. - **Store records:** one business with Canadian defaults, plus the main location. - **Steps:** - Language, clock and owner are already done after start. - The profile can be edited, but `setup_state` can't be changed directly. Unknown steps are refused. - A skipped step becomes one task (skipping twice doesn't make two), and doing it later closes the task. When all steps are done, setup is complete. - **Who may run it:** a manager can't mark steps, and start is refused once the store is set up. A PIN the owner sets for someone is temporary. - **Audit log:** setup writes are logged as `system:setup`, with no secrets.

App

Screen	What
Set up your store (3 parts)	Welcome + language + setup code; clock check against the hub (warns at 2+ minutes); owner name, email, password twice, PIN twice, device name
Recovery code	Shown once, Print button, "I have printed it or written it down" required
Wizard	Progress chips (✓ done, ↻ skipped), resumes at the first unfinished step, every step has Save or Skip
Business profile	Store and legal name, type, address, phone, email, GST/HST, PST, BN, CAD, time zone, fiscal year start, prices with or without tax; the main location gets the same address
Logo and receipt	Logo upload (up to 2 MB), colours from the logo (editable, readability warning), receipt header and footer, live receipt preview
Your team	Add people with role, first PIN (temporary), phone, email
Storage areas	"Add the usual four", add with temperature range, mark not used
Features	Ticked from the shop-type preset, change any; core always on, later ones not available
References	Bank name, processor and merchant ID, CRA payroll, CBSA importer, accountant; warning: no account numbers or passwords
Backups	Explained; skip only (adds the task) until step 7
Home	"Finish setting up your store (N steps left)", task list with "Do it", Store setup tile, "Print a new recovery code" (owner)

Browser check (brand-new local hub, no sample data)

Check	Result
Hub start	<code>setup-code</code> and the monitor message made at startup
First screen	"Set up your store · 1 of 3"; no sideways scrolling at phone width
Wrong code	Refused at the end with a clear message, back to part 1
Clock check	"OK (under 2 minutes)"
Right code typed in lower case without the dash	Owner created; this browser paired as "Owner's computer"; recovery code shown once; Continue locked until confirmed
Business profile	Saved; wizard moved to Logo and receipt (4 of 10)
Logo	Test logo in orange <code>#e8701a</code> and navy <code>#1b2f5e</code> : both found exactly; button colour darkened to <code>#b45714</code> for readable white text; receipt preview with logo, name, address, GST number, footer
Team	Cashier added with a temporary first PIN
Features	Convenience preset ticked Promotions, Labels, Vendors and Purchasing, Regulated Items, Operations
Skips	References and Backups skipped → 8 of 10 done, two tasks on Home with "Do it"
Saved data	Logo file, colours, receipt text, GST number in the database; logo loads without sign-in (200 image/png)

Failed cases and fixes

Symptom	Cause	Fix
---------	-------	-----

Setup routes answered 404	<code>onServe</code> does not exist in PocketBase JS hooks, so the whole file failed to load	<code>onBootstrap</code> (after start-up) plus a once-a-minute job and on-demand from the status endpoint
A <code>sed</code> edit broke a line in <code>setup.js</code>	The inserted comment swallowed the rest of the line	Fixed by hand
Heredoc test script failed in bash	Quotes inside the heredoc	Wrote the test file directly
Svelte warnings "captures the initial value of business"	The forms are meant to start from the saved values once	<code>untrack()</code> makes it explicit
Storage: Shelf showed "0 to 0 °C" and the four areas were in A-Z order	PocketBase number fields have no "empty" (they save 0); all four got the same sort number	0 to 0 shows as "room temperature"; each area gets its own sort number

On the Pi (after deploy)

Check	Result
Deploy	Backup integrity ok; <code>1791200007_p0_setup.js</code> applied; monitor units installed; no errors
Monitor message	<code>console.issue = "Chedam hub: https://4"</code> (this dev hub has an owner, so no setup code); <code>/etc/issue.d/chedam.issue</code> linked; <code>chedam-console.path</code> active
<code>/api/chedam/setup/status</code>	<code>needs_setup: false</code>
PocketBase memory	41 MB

The setup code on the **monitor** will be seen for real on a brand-new hub, for example the spare-card restore test in step 7/10. The dev hub keeps its sample store.

15. P0 step 7: backups (FR-1.14, NFR-03, NFR-04, NFR-12)

Sreya's plan: use her hard drive as the second drive in the testing phase. Production: OTG adapter plus a spare SD card.

Design (DL-48 to DL-52)

Part	What
Root helper <code>hub/system/chedam-backup</code> (Python, installed to <code>/opt/chedam/bin</code>)	Actions: <code>drives</code> , <code>key-init</code> , <code>key-show</code> , <code>run</code> , <code>test-restore</code> ; technician-only <code>restore</code> <code>FILE --key KEYFILE</code>
Hub ↔ helper	Hub writes <code>pb_data/backup/request.json</code> ; <code>chedam-backup.path</code> runs the helper (low priority: <code>Nice=10</code> , idle I/O); answer in <code>result.json</code>
One backup	SQLite online snapshot → <code>PRAGMA integrity_check</code> → <code>gzip</code> → age encryption → <code>Chedam-Backups/<store>/chedam-YYYYMMDD-HHMMSS-<kind>.db.gz.age + .sha256 + .ca</code> (HTTPS CA, encrypted) → read back from the drive, decrypt, check again, compare table counts = verified
Retention	Newest per day 14, per week 8, per month 12, plus the newest 3; only Chedam's own files
Key	<code>age X25519</code> key in <code>/opt/chedam/keys</code> (root only); owner prints it; owner can show it again
Hub (<code>lib/backup.js</code> , <code>backups.pb.js</code> , <code>migration ... 08_p0_backups</code>)	Status, <code>drives</code> , choose drive, key (owner), <code>run</code> , restore test (owner), nightly schedule (<code>backup.schedule</code> , retry after 1 h, 3 per day), lost-job detection (30 min), one urgent task on failure closed by the next good backup, first backup completes the setup step
App	Wizard Backups step: drive → key (print, confirm) → first backup. Backups screen: last good backup, back up now, restore test, drive and nightly time, backup key, recent backups. Home tile
Scripts	<code>deploy.sh</code> and <code>setup-hub.sh</code> install <code>age</code> , the helper and its units

Sreya's hard drive

Drive	Seagate ST31500341AS (Barracuda 7200.11, about 2009), 1.5 TB, NTFS, label "SCRAPP", in a USB enclosure (XT-U33502) with its own power; 1.07 TB free
Pi power	Clean the whole time (<code>throttled=0x0</code>); a 3.5" drive has its own adapter
Problem	4 read errors, each after a 30 s timeout and a USB reset, at sectors 6,229,920 to 6,230,944 (about 3.2 GB into the disk). On NTFS that is near where the Master File Table usually sits, and writing any file updates that table
Decision	Nothing was written to the drive , to protect Sreya's files. It was only mounted read-only once (to read free space) and unmounted. Advice: copy anything important off it on Windows and check it there (<code>chkdsk</code>); use a healthy USB stick or drive for Chedam backups

Failed cases and fixes

Symptom	Cause	Fix
Restore test got 409 "A backup job is still running"	A job marked failed as "lost" (no answer for 30 min) left the in-memory busy flag set	The flag is cleared with the lost job, and a run counts as busy only while its record is still running (test caught it)
Test "request went to the helper" failed	It checked before the fake helper's 100 ms poll	Wait 400 ms
Listing drives took 2 minutes with the hard drive	The listing mounted each drive to read free space; the sick drive took minutes to mount	The listing never mounts (now 0.55 s); free space comes from each backup
A stuck mount could hold up the helper indefinitely	No time limit	90 s limit; reports "The drive is not responding"
The test image was not listed	Loop devices have type "loop"	Allowed for test images only (<code>CHEDAM_BACKUP_ALLOW_LOOP=1</code>)
Test script lost the image ID	<code>set -e</code> stopped before saving it; <code>losetup</code> needs <code>sudo</code>	Looked up by file
The pre-restore copy was readable by every account on the Pi	<code>copy2</code> kept root 644	<code>chedam-hub</code> , 0640
A rebuilt hub would get a new HTTPS certificate authority (every phone reinstalls)	Not part of the backup	Each backup saves the CA (encrypted); restore puts it back
step2 test failed once: "unknown shorthand flag: 'O'"	The random test password started with "-" (about 1 in 64) and was read as a command option	Test passwords start with "Pw" (both harnesses)

Tests

PC: `node hub/tests/run-all.mjs` → **281** (step7-backups 30/30, with a fake helper that answers like the real one).

On the Pi, the real helper with a 256 MB disk image (ext4) as the drive:

Check	Result
Drive list	0.55 s; lists the image and Sreya's drive without mounting
Key	Made; secret never printed in the terminal
Backup	1.1 s , 44 KB encrypted; verified (read back, decrypted, integrity ok, same counts: 5 users, 100 events, ...)
On the drive	Only <code>Chedam-Backups/demostore/...age</code> , <code>.sha256</code> , <code>.ca</code> ; file starts with <code>age-encryption.org/v1:sha256sum -c OK</code>
Restore test	Consistent, 0.2 s
Real restore (<code>chedam-backup restore ... --yes</code>)	2-3 s including the hub restart; health ok; integrity ok; same data; pre-restore copy kept

HTTPS after restore	Same CA fingerprint (12:4B:3C:BF...); the laptop with this morning's <code>ca.crt</code> connects without change
Hub-to-helper path	A request written as <code>chedam-hub</code> was picked up by the path unit and answered
Power	<code>throttled=0x0</code> throughout; PocketBase 39 MB

Restore procedure for a dead hub: `docs/P0-backup-restore.md`.

16. Backup card, sign-in mix-up, certificates, 3 devices signed in

New backup card

Sreya connected a 7.4 GB SD card in a USB reader ("Storage Device", FAT32, UUID A8D7-32C4). Read-only check: the first 64 MB read in 5 s with no errors. Power stayed clean (`throttled=0x0`). The card is supported; nothing has been written to it yet. The first backup is made from the app (Backups → drive → key → Back up now → Test a restore).

Sign-in mix-up (not a logic bug)

What Sreya saw	Cause (from the hub's records)	Fix
iPhone at <code>https://chedam.local</code> showed a "registration" page	The iPhone was not paired : its pairing code ("13pro") was made but never used, so the app showed "Pair this device"	New pairing code from the owner on the laptop, scanned with the iPhone
Owner email + password refused on the iPhone and the laptop	Demo Owner never had a password : the dev sample store gives every person a random, unknown password (0 password changes in the audit log). The file <code>pb_migrations_dev/1791200101_dev_sample_pins.js</code> holds PINs for the name + PIN screen, not passwords	Sreya set a password in the admin UI (users → Demo Owner)
"Locked"	5 wrong tries locked the owner for 15 minutes (<code>pin_locked_until</code> 00:53 UTC), the lockout working as designed (NFR-11)	Cleared <code>pin_locked_until</code> in the admin UI (or wait 15 min)
Sample people had wrong tries (Demo Owner 2, Cal Cashier 4)	Guessed PINs	Use the PINs from the dev seed file

Lessons for the docs and for later: - The dev sample owner needs a known way in. A dev hub's owner password is set in the admin UI (`/_/_`). The setup wizard does this properly for real stores. - Telling apart "not paired", "no password set" and "locked" is hard for the user. The screens could be clearer (for example "This device is not paired yet" in place of a page that looks like registration). Noted for the step 5/6 polish.

Certificates on the three devices

Device	Before	After
Android 9 (9lite)	Trusted (paired, HTTPS since the morning)	Same
Surface laptop (Windows, Chrome)	Not trusted : Chrome showed "Not secure" with <code>https</code> crossed out on <code>https://192.168.50.101/_/_</code>	Sreya installed <code>ca.crt</code> into Current User → Trusted Root Certification Authorities and restarted Chrome
iPhone 13 Pro (iOS 26.6.1, Chrome 154)	Profile installed; trust toggle needed checking	Certificate Trust Settings → Chedam Store CA on; paired

Test for any device: `https://192.168.50.101/hub.json` opens with no warning.

Result: 3 devices signed in (P0 gate item "3 devices log in")

Device	Hub record	Browser (from the hub)	Signed in
iPhone 13 Pro	"13pro", approved, app 0.5.0	iOS 26.6.1, Chrome 154 (CriOS)	Demo Owner

Android 9 phone (Honor 9 Lite, LLD-AL10)	"9lite", approved, app 0.5.0	Android 9, Chrome 98	Demo Owner
Surface laptop	Owner on any device (not paired, FR-1.09)	Chrome on Windows	Demo Owner

Two earlier iPhone pairing attempts ("13pro", "13pronew") were removed in the device manager (status revoked), as intended.

Issue: Android 9 phone shows misaligned screens

Symptom	Screens work but are not properly aligned on the Android 9 phone
Cause	The phone's Chrome is version 98 (February 2022). The app uses Tailwind CSS 4, which needs Chrome 111+ (cascade layers, <code>color-mix()</code> , <code>oklch</code> colours). Chrome 98 ignores those rules, so part of the styling is lost. My assumption that Android 9 would have Chrome 138 was wrong (risk note in the mini-spec)
Options	(a) Update Chrome on that phone from the Play Store (Android 9 still gets Chrome up to about 138), which fixes it at once if updates work. (b) Make the app's CSS work on older Chrome too (Tailwind 3.4 or a compatibility build), the safer choice for pilot stores with old phones
Status	Decided (DL-53): no extra work; Sreya will test on another, up-to-date phone

17. P0 step 8: hub health page (FR-12.09)

What it shows

Item	Source (read as the hub's sandboxed user)	Status rule
Hub running for	<code>/proc/uptime</code> , model from <code>/proc/device-tree/model</code>	info
Temperature	<code>/sys/class/thermal/thermal_zone0/temp</code>	warn ≥ 70 °C, bad ≥ 80 °C (the Pi slows down)
Power supply	<code>rpi_volt</code> sensor <code>in0_lcrit_alarm</code> (under-voltage now)	bad if too low
Hub app memory	<code>/proc/self/status</code> <code>VmRSS</code>	warn > 150 MB (NFR-08)
Free memory	<code>/proc/meminfo</code> <code>MemAvailable</code>	warn < 100 MB, bad < 50 MB
Free storage / store data	<code>df</code> on <code>pb_data</code> ; size of <code>data.db</code> + WAL	warn < 15 %, bad < 5 %
Clock	<code>chronyc -c tracking</code> (stratum, offset, leap); <code>/dev/rtc0</code>	warn if not synchronised or > 2 s off; notes "no battery clock (RTC)" (DL-23)
Internet	connectivity status (FR-12.01)	info
Last backup	<code>backups</code> status	bad if no drive or > 50 h; warn > 26 h (NFR-04)
Last restore test	<code>backup.last_restore_test</code>	bad if failed; warn > 35 days (NFR-03)
Devices online	device manager view	warn if devices wait for approval
Waiting to sync	offline queue arrives with selling (P1)	info
Versions	app build (<code>sw.js</code>), schema (newest non-dev migration), PocketBase, Caddy, OS, kernel	info
Pending updates	<code>updates</code> table (channel in step 9)	warn if one failed

- The top line shows the worst status ("All good" / "Needs a look" / "Needs attention now"). Answers are cached for 20 s, and "Check now" forces a fresh check.
- Endpoint: `GET /api/chedam/health` (health.view: owner, manager). App: Hub health screen (refreshes every 30 s, links to Backups and Devices), a home tile with ✓ / ! / ✗, and a red "The hub needs attention" card on the home screen.

Checks

Check	Result
-------	--------

Which readings the hub user can access on the Pi	All, except <code>vcgencmd/throttling</code> flags (needs root); the <code>rpi_volt</code> sensor gives under-voltage instead
Same commands inside the hub's systemd sandbox (<code>systemd-run</code> with the service's restrictions)	chrony: synced, stratum 2; df: 117.7 GB free of 125.4 GB; 40.8 °C; power alarm 0; Caddy 2.6.2; PocketBase 0.40.4
Tests <code>step8-health.test.mjs</code>	21/21: 401 signed out, 403 cashier, every item present with a status, no backup drive → bad, 30 h old backup → warn, fresh → ok, restore test "never", schema shown, cached answers
All tests	302 passing
Deploy	Backup integrity ok; no errors; endpoint answers 401 without sign-in

Failed cases and fixes

Symptom	Cause	Fix
"Schema" showed <code>1791200101_dev_sample_pins</code>	The newest migration was the dev sample-data one	Ignore files containing <code>_dev_</code>
The fix then showed "Schema ?"	Backslash escaping through Python → JS → SQL <code>ESCAPE</code> broke the query (the same trap as on day 2)	<code>instr(file, '_dev_') = 0</code> , no escaping needed

18. P0 step 9: updates (FR-12.02, FR-12.04, NFR-20)

Sreya created the public repo `repatikosamid-prog/chedam-releases` (DL-54). The full guide is in `docs/P0-updates.md`; the decisions are DL-55 to DL-58.

What was built

Part	What
Signing key	Ed25519 made with OpenSSL in <code>~/.chedam/update-signing.key</code> (outside the repo; Windows permissions: Sreya's account only); public key <code>hub/system/update-signing.pub</code>
<code>tools/release/make-release.sh</code>	Build the app → stage hub code (refuses dev migrations) → <code>manifest.json</code> (version, notes, commit, migrations) → <code>tar.gz</code> → sign → check the signature against the repo's public key → signed <code>stable/channel.json</code> in <code>../chedam-releases</code> ; <code>--publish</code> pushes
<code>hub/system/chedam-update (root) + .path/.service</code>	<code>check</code> (https channel, signatures, checksum), <code>scan</code> (USB <code>chedam-update/</code> and <code>/opt/chedam/updates/inbox</code>), <code>install</code> (signature → DB snapshot + code copy → swap → restart → health or automatic rollback)
Hub: migration ... <code>09_p0_updates</code> , <code>lib/updates.js</code> , <code>updates.pb.js</code>	Status, check online/USB, install now (owner) or in the 02:00–05:00 window, results read after the restart, an urgent task on failure, a daily background check when online (not blocking)
App	Updates screen (installed version, check online / USB stick, release notes, "signed ✓", install now / tonight, history), home tile; product version 0.9.0
Scripts	<code>deploy.sh/setup-hub.sh</code> install the helper, units and public key; <code>/opt/chedam/keys</code> stays root-only (it also holds the backup key); a deploy writes <code>VERSION = dev-<commit></code>
Local clone	<code>C:\Users\Venkata\Desktop\Projects\chedam-releases</code> with a README

Tests

PC: `step9-updates.test.mjs` 21/21 with a fake helper: - **Check**: cashier refused; online check finds 2 updates; the channel address is passed; no duplicate records; the offline check is recorded. - **USB scan**: finds the package. - **Install**: manager refused, owner installs now; a second install is refused while one runs (409); the result is read after the restart. - **Rollback**: reported, with an urgent task. - **Install window**: scheduled → installed when the window opens; the task closes. - **Daily check**: sent without waiting, its answer recorded at the next tick. - **Audit log**: installs and rollbacks are logged.

All suites: **323**.

On the Pi with the real helper and real signed packages (local, not yet published):

Test	Result
Package 0.9.0	118,545 bytes, 44 files, no dev data, keys or databases; signature ok
Scan of the inbox	Found 0.9.0 (signed)
Install 0.9.0 over <code>dev-cd633e3</code>	1.7 s (3 s end to end); VERSION 0.9.0; app build <code>0.9.0-15ad57cd6968</code> ; 5 users and 129 events unchanged; integrity ok; dev migrations kept
Broken 0.9.1 (signed; migration throws on purpose; built in the scratch folder only)	First run: rolled back after 93 s (waited out the 90 s limit). After the fail-fast fix: rolled back after 4 s ; version 0.9.0; data and integrity unchanged; broken migration never applied
Good 0.9.0 again (<code>--force</code>) after the fixes	1.6 s, healthy
Tampered package (1 byte added)	Refused: "The signature does not match"; nothing changed

Failed cases and fixes

Symptom	Cause	Fix
Permission command for the key failed (<code>Invalid parameter ":USERNAME:(R,W)"</code>)	PowerShell variables written inside a Bash command	Ran it in PowerShell; the key is now readable by Sreya's account only
<code>deploy.sh/setup-hub.sh</code> would have opened <code>/opt/chedam/keys</code> to 755	Copy-paste of the updates folder line	Kept at 700 (the backup key lives there)
Hub test: second install was not refused; test crashed	The fake helper finished in 0.2 s, so the first install was already done	The fake install takes 1.5 s; the 409 is checked at once
Daily check would block a hub job for up to 2 min	It waited for the helper's answer	It now only sends the request; the answer is read at the next tick
<code>make-release.sh</code> : "Cannot find module 'c/Users/.../package.json'"	Node on Windows cannot read Git Bash paths written inside a script	Read the version with a relative path
Building the broken test package: "Cannot connect to C: resolve failed"	GNU tar takes <code>C:</code> as a remote host	Used <code>/c/...</code> paths
Broken update took 93 s to roll back	The installer waited the full 90 s although the hub logged "failed to apply migration" after 3 s	It watches the log while waiting and rolls back at the first crash
After that fix the rollback was reported as "did not come up" although it had worked	The rollback check read the log "since this second", which still held the crash lines from the same second	Journal cursors : each check reads only lines written after it started

19. First real backup verified; 0.9.0 published

Backup to the new SD card (made by Sreya in the app)

Check	Result
Records	2 backups, both verified : a nightly one, which started by itself the moment the drive and key were ready (tonight's slot had not run yet, the catch-up rule), and the first backup from the wizard (48,118 bytes); 7.9 GB free
Setup task "choose a backup USB drive"	done
Sreya's restore test in the app	passed, consistent, 0.3 s
Independent check (card mounted read-only)	Only Chedam's folder added (plus Windows' "System Volume Information"); <code>sha256sum -c</code> OK for both; both files start with <code>age-encryption.org/v1;</code> <code>.ca</code> files present
Independent restore test (helper)	Newest backup rebuilt in 0.3 s, consistent (5 users, 4 devices, 120 events, ...)
Power	<code>throttled=0x0</code> (the 4 I/O errors on record are the old hard drive's)

Publishing 0.9.0 (with Sreya's OK)

```
bash tools/release/make-release.sh --notes "Chedam 0.9.0 (P0): ..." --publish
```

Check	Result
Package	chedam-app-0.9.0.tar.gz, 118,582 bytes, sha256 d6f0b698..., built from committed code, signature ok
Public repo chedam-releases	Commit 3c87ec6: README, stable/channel.json(.sig), packages/chedam-app-0.9.0.tar.gz(.sig)
Pi downloads from GitHub (chedam-update check)	Channel and package signatures valid, checksum matches; 0.9.0 found as newer than dev-f632959

Issue found and fixed

Symptom	Cause	Fix
Git warned "LF will be replaced by CRLF" for files in the releases repo	Git on Windows may convert line endings in files it takes for text. A 64-byte signature has no zero byte about 78% of the time, so Git can treat it as text, and changing a single byte breaks the signature. This time the files arrived intact (verified by the Pi)	.gitattributes with * -text in chedam-releases (commit ba25f3c); make-release.sh refuses to run without it

20. P0 gate: memory under load (NFR-07, NFR-08, risk R6)

Method

Part	What
Test hub	tools/loadtest/pi-loadtest.sh: a throwaway hub on the Pi (same PocketBase binary, hooks and migrations including the sample store; its own data folder; 127.0.0.1:8099 only), started with the same limits as the service (GOMEMLIMIT 100 MiB, MemoryHigh 150 MB, MemoryMax 220 MB), plus a sampler (RSS/peak every 2 s, load, free memory). The store's real hub was not touched. Deleted afterwards
Load	tools/loadtest/hub-load.mjs on the laptop through an SSH tunnel: 10 devices (twice the pilot's 5, NFR-07), each paired and signed in by PIN, every 1-2 s: connectivity, device status, my permissions, settings, modules, storage areas; writes (create + update, each also writes an event) by the roles allowed to; a PIN sign-in every 10 rounds (bcrypt); health page; plus every 30 s a burst of simultaneous audit-log reads (100 events per page)
Duration	5 minutes per run

Results

Run	Requests	Errors	Latency p50 / p95 / p99	Hub memory peak (VmHWM)	Pi free memory (lowest)	NFR-08
1 (script mistakes, see below)	12,729	1,106	34 / 136 / 530 ms	86 MB	174 MB	(not valid)
2: burst of 50	10,805	0	37 / 766 / 5,034 ms	197 MB	92 MB	FAIL
3: burst of 10 (all devices at once: realistic)	11,723	0	35 / 94 / 552 ms	108 MB	174 MB	PASS
4: burst of 50, with rate limits	11,961	360 × 429 (bursts only)	36 / 470 / 920 ms	120 MB	162 MB	PASS

The real hub afterwards: 43 MB, healthy through Caddy.

Fix: per-device rate limits (DL-59)

- Migration 1791200010_p0_rate_limits.js:
- limits: audit-log lists 10 per 3 s, all lists 40 per 3 s, PIN sign-in 15 per 10 s, all API calls 150 per 3 s;
- the device's address comes from X-Forwarded-For (Caddy sets it), and 127.0.0.1 is exempt.
- Without the trusted header, every device would look like 127.0.0.1 behind Caddy and share one allowance.

- Test `step10-ratelimit.test.mjs` (5/5):
- a flooding device gets about 10 answers, the rest 429;
- another device is not affected, and the flooder is served again after 3 s;
- a busy till (36 requests in 3 s) is never limited;
- 127.0.0.1 is exempt.
- All suites: **328**.

Failed cases and fixes

Symptom	Cause	Fix
Run 1: 1,106 errors	Test script: cashier and accountant tills tried to add storage areas (correctly refused, 403); the bursts sorted the audit log by <code>created_at</code> , but its field is <code>at</code> (400)	Writes only by roles allowed to; sort by <code>at</code> . The run-1 memory figure was not valid: the bursts had failed instantly
Run 2: 197 MB, p99 5 s	50 simultaneous audit-log reads: each SQLite connection keeps its own page cache, and PocketBase has no option to cap connections	Per-device rate limits (DL-59)
"Through Caddy" check on the Pi failed (curl exit 77)	My Pi account cannot read Caddy's certificate file	Checked from the laptop: 200

Next

1. Sreya: install 0.9.0 from the Updates screen (steps in the chat).
2. Gate tests left: **power pull ×10** mid-write (Sreya pulls the plug, I run the writes and the checks) and **camera over HTTPS** on the iPhone and an up-to-date Android phone.

2026-10-03 (Saturday, October 3, 2026)

New network: own router, Pi re-flashed, SSH keys documented, HDMI fixed

Source: docs/P0-pi-setup-log-2026-10-03.pdf (detailed log of this day) and docs/P0-restart-new-network.md (the plan). This entry carries its content forward.

Summary

- The TELUS modem gives no admin access, so DHCP reservations are impossible. The dev hub moved behind our **own TP-Link router**, which is LAN-cabled to the bedroom wall port.
- Network: subnet 192.168.50.0/24, gateway 192.168.50.1, 2.4 GHz SSID TP-LINK_2CBE.
- Pi chedam at **192.168.50.101** (MAC 88:A2:9E:5C:17:65, reservation enabled); dev PC DESKTOP-AHG4TD7 at 192.168.50.100.
- The SD card was wiped and re-flashed. Key-only SSH works, and the OS is updated.
- HDMI console made visible. setup-hub.sh gained the --keep-display option.
- Commits: ea0a7e6, 678eee4, bacf585.

Why SSH keys instead of a password

With a key pair, the private key never leaves the PC and the Pi holds only the public key. No secret crosses the network, keys cannot be guessed, and scripts can log in, so password login is switched off.

File	Where	Rule
id_ed25519	PC C:\Users\Venkata\.ssh\	Private. Never share, paste or commit; back it up privately
id_ed25519.pub	PC, same folder	Public, one line ending chedam-dev. Safe to copy
authorized_keys	Pi /home/chedam/.ssh/	Allowed public keys; folder 700, file 600
known_hosts	PC .ssh\	Remembered Pi identities; ssh-keygen -R <ip> after a re-flash
config	PC .ssh\	Alias ssh chedam

Key procedures recorded:

- create the key pair
- put the public key on the Pi with Imager (REMOVE old keys, then BROWSE to the .pub file)
- add or replace keys with type new.pub | ssh chedam "cat >> ~/.ssh/authorized_keys"
- what to do when locked out (monitor + keyboard, or re-flash)

Steps

1. Backed up .ssh with Copy-Item \$env:USERPROFILE\.ssh \$env:USERPROFILE\.ssh_backup -Recurse.
2. Cleared old host fingerprints by deleting known_hosts and known_hosts.old. There was a backup.
3. Router: changed the admin password, set the LAN to 192.168.50.1/24 (different from TELUS 192.168.1.x), kept it in router mode, and turned on 2.4 GHz.
4. Re-flashed the card. Imager: Pi Zero 2 W, Lite 64-bit, chedam/chedam, new Wi-Fi, CA, SSH public-key only.
5. Found the Pi in the router's DHCP client list and created the reservation.
6. Updated the SSH config to HostName 192.168.50.101 and removed HostKeyAlias.
7. Ran sudo apt update && sudo apt full-upgrade -y and checked that sshd -T shows passwordauthentication no.
8. Readiness check: model, OS, IP, active Wi-Fi, multi-user.target; reboot test passed.
9. Forced HDMI output (see below).

Failed cases and fixes

Attempt / symptom	Cause	Fix
ssh-keygen -R <old-ip> errored	PowerShell treats < as an operator	Use real addresses; deleting known_hosts avoided needing the old IP
Key pair deleted during "start fresh"	Not needed: a key is not tied to a network	Restored from .ssh_backup
ssh 192.168.50.101 denied as venkata	No user given, so the Windows name was used	ssh chedam@192.168.50.101
ssh chedam@chedam timed out	Windows cannot resolve a bare name	IP, chedam.local, or the alias
-i \$env:venkata\.ssh\... not found	\$env:USERPROFILE is a real variable, not a placeholder	Keep \$env:USERPROFILE
Permission denied (publickey)	Imager wrote an old remembered key (... COL3o srey...), not the PC's (... LuhpI0dY chedam-dev)	Re-flash; REMOVE old keys, BROWSE to id_ed25519.pub
A command pasted into Imager's key box	Get-Content ... \ Set-Clipboard text pasted as the key	REMOVE, then BROWSE to the file
"REMOTE HOST IDENTIFICATION HAS CHANGED"	Expected after a re-flash (new host key)	ssh-keygen -R 192.168.50.101
Reservation saved as Disabled (TP-Link)	Router default	Tick it, then Enable Selected; leave ARP binding off
HDMI blank	dmesg: vc4-drm: Cannot find any crtc or sizes	Append video=HDMI-A-1:1280x720@60D to the single line in /boot/firmware/cmdline.txt, then reboot
Lines after sudo cp lost	The sudo password prompt ate the pasted lines	Paste one command at a time when sudo may prompt

HDMI fix commands:

```
sudo cp /boot/firmware/cmdline.txt /boot/firmware/cmdline.txt.bak
sudo sed -i '1 s/$/ video=HDMI-A-1:1280x720@60D/' /boot/firmware/cmdline.txt
sudo reboot
```

Lessons

- Before flashing, check that the key shown in Imager ends the same as type C: \Users\Venkata\.ssh\id_ed25519.pub. Imager remembers old keys.
- Always back up .ssh before changes.
- With password login off, losing id_ed25519 means re-flashing or using a keyboard and screen.
- No passwords or key contents are stored in the repo.

2026-10-02 (Friday, October 2, 2026)

P0 kickoff: spec review, dev PC tools, first Pi flash and first hub setup (TELUS network)

Summary

- Read Master Specification v1.0 and pulled out the P0 (Foundation) scope, done-criteria and phase gate.
- Wrote `docs/P0-readiness-checklist.md`: hardware, software with versions and links, Pi flashing steps, accounts.
- Installed and verified the developer tools on the Windows PC (all checksums matched the official ones).
- Flashed the Pi. The first attempts failed on SSH keys (including the private key being pasted into Imager); the key was replaced and SSH worked.
- Wrote the hub setup (`hub/scripts/setup-hub.sh` + `hub/system/*`) and ran it on the Pi at `192.168.1.241` (TELUS network). HTTPS worked by name and IP.
- Wrote `docs/P0-mini-spec.md` (P0 build order, decisions DL-21 to DL-25).
- Decisions from Sreya: RTC skipped for now; cloud backup tested later when client IDs exist; test phones are Android 9 and iPhone 13 Pro (iOS 26.6.1); update channel option A (source stays private, signed packages in a separate public repo).

1. Spec review

Source: `SDLC Docs Tools software installers and ReadMe/MASTER Specifications/Chedam - Master Specification v1.0.pdf` (41 pages).

The PDF text was extracted with `pdftotext -layout`. The release-plan and architecture pages are images, so they were rendered with PyMuPDF and read as pictures.

P0 requirements: FR-1.01 to 1.10, 1.13, 1.14, 1.15; FR-12.01 to 12.04, 12.08, 12.09.

Done when: 3 devices log in, the hub survives a power pull, an update installs and rolls back. The phase gate adds: memory under 150 MB (NFR-08), a load test (R6), an iOS/Android HTTPS camera test (R5), and a verified backup and restore.

2. Developer PC tools

Already present: Git 2.52.0, VS Code 1.135, Python 3.13, Windows OpenSSH.

Tool	Version	Verified against	Result
Node.js	24.21.0 (LTS Krypton)	nodejs.org SHASUMS256 + OpenJS signature	Match, installed
PocketBase (Windows)	0.40.4	PocketBase <code>checksums.txt</code>	Match
DuckDB CLI	1.5.6	GitHub release digest	Match
Raspberry Pi Imager	2.0.11.1	GitHub digest + Raspberry Pi Ltd signature	Match, installed
DB Browser for SQLite	3.13.1	winget manifest + SignPath signature	Match, installed
mkcert	1.4.4	Scoop manifest (file is unsigned)	Match
GitHub CLI	2.102.0	n/a	Installed, logged in as <code>repatikosamid-prog</code>

Commands used for verification (Git Bash):

```
sha256sum *.msi *.exe *.zip
curl -s https://nodejs.org/dist/v24.21.0/SHASUMS256.txt | grep x64.msi
```

```
powershell Get-AuthenticodeSignature <file>
```

Failed case: `mkcert` not recognized

- Symptom: `mkcert -install` → "The term 'mkcert' is not recognized".
- Cause: the file was still named `mkcert-v1.4.4-windows-amd64.exe`, and `C:\Tools` was not on PATH. PowerShell also never runs a program from the current folder without `.\`.
- Fix: copied `mkcert.exe`, `pocketbase.exe` and `duckdb.exe` to the top of `C:\Tools` and added `C:\Tools` to the user PATH. A new PowerShell window was needed for the change to show.

Then Sreya ran `mkcert -install` (local dev CA added to the Windows trust store) and `gh auth login` (fine-grained token).

The installers folder was added to `.gitignore`. PocketBase had created test data there.

3. Flashing the Pi (first time)

Imager settings: Pi Zero 2 W, Raspberry Pi OS Lite 64-bit, hostname `chedam`, user `chedam`, Wi-Fi 2.4 GHz, country CA, time zone America/Vancouver, SSH public-key only.

Failed case 1: hostname and username

- The hostname was typed as `Chedam` and the username as the GitHub name.
- Fix: lowercase `chedam` for both, so every script and guide matches.

Failed case 2: no SSH key existed

- SSH was set to password authentication, and the PC had no key yet.
- Fix: `ssh-keygen -t ed25519 -C "chedam-dev"`.

Failed case 3: `Permission denied (publickey)`

- Debug output showed the PC offering its key and the Pi refusing it for both usernames:

```
ssh -v -o BatchMode=yes chedam@chedam.local true
```

- Cause: the key written to the card was not the PC's key.
- The "host key differs" warning came from an old `192.168.1.241` entry in `known_hosts`. It was removed with `ssh-keygen -R`.

Failed case 4 (security): private key pasted into Imager

- Imager listed 7 "keys". They were the 7 lines of the **private key** (`BEGIN OPENSSH PRIVATE KEY`).
- Risk: the private key was copied onto the SD card and partly shown in a screenshot.
- Fix: removed all 7 entries, deleted the key pair, generated a new one, and pasted **only** `id_ed25519.pub` (one line, `ssh-ed25519 ... chedam-dev`). Re-flashing overwrote the card.

Failed case 5: `.local` name not resolving from Windows tools

- Git Bash `ssh` and Windows `ssh` both failed on `chedam.local`, while `Resolve-DnsName` worked.
- Fix: an SSH alias `chedam` pointing at the IP, in `~/.ssh/config`. The Pi first booted slowly (~5 min), so early checks also failed.

Login worked: `chedam@chedam`, Debian 13 trixie, aarch64, 415 MB RAM, 117 GB card.

Admin access

`sudo` asked for a password, so Sreya ran this once:

```
ssh -t chedam "echo 'chedam ALL=(ALL) NOPASSWD:ALL' | sudo tee /etc/sudoers.d/010-chedam && sudo chmo
```

4. Hub setup (first time, Pi at 192.168.1.241)

OS update (ran in the background, ~15 min):

```
sudo apt-get update && sudo apt-get -y full-upgrade && sudo apt-get -y install chrony sqlite3 ufw
```

PocketBase 0.40.4 linux_arm64 was downloaded on the Pi and checked with `sha256sum -c` against `checksums.txt`.

Files written in the repo:

File	Purpose
hub/scripts/setup-hub.sh	Idempotent setup: packages, service user <code>chedam-hub</code> , <code>/opt/chedam</code> , PocketBase service, Caddy, watchdog, journald, Wi-Fi power save off, boot config, firewall
hub/system/chedam-hub.service	PocketBase on 127.0.0.1:8090, auto-restart, <code>GOMEMLIMIT=100MiB</code> , <code>MemoryMax=220M</code> , hardening
hub/system/Caddyfile	HTTPS for <code>chedam.local</code> and the hub IP, using a per-store CA; plain HTTP only for <code>/ca.crt</code>
hub/system/watchdog.conf	Hardware watchdog 14 s
hub/system/journald.conf	Logs capped at 64 MB (SD card wear)
hub/system/wifi-powersave-off.conf	Lower Wi-Fi latency
hub/system/index.html	Placeholder page "Chedam hub is running"

Run on the Pi:

```
scp -r scripts system chedam:hub/
ssh chedam 'sudo bash ~/hub/scripts/setup-hub.sh'
ssh chedam 'sudo systemctl reboot'
```

Failed case 6: GitHub repo creation

- `gh repo create ... --public` → "Resource not accessible by personal access token".
- Cause: the fine-grained token cannot create repositories. Sreya has to create the public repo on github.com.

Failed case 7: HTTPS tests from the PC returned nothing

- Windows `curl` (Schannel) → "the revocation status is unknown". A private CA has no revocation server.
- Fix for command-line tests: `--ssl-no-revoke`. Browsers are not affected.

Failed case 8: Caddy error at start

- Caddy tried to install its root certificate into the Pi's own trust store and failed ("failed to execute sudo").
- Fix: the `skip_install_trust` global option in the Caddyfile.

Results after reboot

Check	Result
Services	chedam-hub, caddy, chrony, avahi-daemon active
HTTPS	<code>https://chedam.local/api/health</code> and <code>https://192.168.1.241/api/health</code> → 200
Certificate	Valid against "Chedam Store CA - 2026 ECC Root" (valid to 2036); IP cert has IP SAN
Memory	PocketBase 27 MB + Caddy 41 MB; total RAM rose from 415 to 462 MB after headless config
Watchdog	14 s
Temperature	52.6 °C, no throttling

5. Decisions recorded (docs/P0-mini-spec.md)

ID	Decision
DL-21	HTTPS via Caddy with an internal CA created on each hub (one per store); mkcert is dev-PC only
DL-22	Source repo private; signed update packages in a separate public repo
DL-23	RTC deferred for dev; required for the pilot
DL-24	Headless boot config (later changed to a switch, see 2026-10-05)
DL-25	PocketBase extended with JavaScript hooks; no Go build

Findings:

- Android 9 cannot resolve `.local`, so the hub needs a fixed IP.
- Android 9's Chrome no longer gets updates.
- Caddy leaf certificates last 12 hours, so the clock must be correct (RTC for the pilot).